

Calculation of No-Wall Vacuum Matrix

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Let

$$\mathcal{E}_m(r, \theta, \phi) = \bar{\mathcal{E}}_m(r, \theta) e^{in\phi}, \quad (1)$$

where

$$\mathcal{E}_m(\epsilon, \theta, \phi) = e^{-i(m\theta - n\phi)}, \quad (2)$$

$$\mathcal{E}_m(\infty, \theta, \phi) = 0, \quad (3)$$

$$\nabla^2 \mathcal{E}_m = 0. \quad (4)$$

It follows that

$$\bar{\mathcal{E}}_m(\epsilon, \theta) = e^{-im\theta}, \quad (5)$$

$$\bar{\mathcal{E}}_m(\infty, \theta) = 0, \quad (6)$$

$$\nabla^2 \bar{\mathcal{E}}_m = \frac{n^2}{R^2} \bar{\mathcal{E}}_m. \quad (7)$$

We can ensure that Eqs. (3) and (4) are satisfied by writing

$$\mathcal{E}_m(r, \theta, \phi) = \oint \oint_{r'=r_c} \frac{\sigma_m(\theta') e^{in\phi'}}{|\mathbf{r} - \mathbf{r}'|} \mathcal{J} d\theta' d\phi', \quad (8)$$

where $r_c < \epsilon$, It follows that

$$\mathcal{E}_m(r, \theta, \phi) = e^{in\phi} \oint \oint_{r'=r_c} \frac{\sigma_m(\theta') e^{in\phi''} r' R'^2}{[R^2 + R'^2 - 2 R R' \cos \phi'' + (Z - Z')^2]^{1/2}} d\theta' d\phi'', \quad (9)$$

where $\phi'' = \phi' - \phi$, which reduces to

$$\bar{\mathcal{E}}_m(r, \theta) = \oint \oint_{r'=r_c} \frac{\sigma_m(\theta') \cos(n\phi) r' R'^2}{[R^2 + R'^2 - 2 R R' \cos \phi + (Z - Z')^2]^{1/2}} d\theta' d\phi. \quad (10)$$

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However,

$$P_{-1/2}^n(\cosh \eta) = \frac{(-1)^n \Gamma(1/2) \Gamma(1/2 + n)}{2\pi^2} \oint \frac{\cos(n\phi) d\phi}{(\cosh \eta - \sinh \eta \cos \phi)^{1/2}}. \quad (11)$$

Hence, we get

$$\bar{\mathcal{E}}_m(r, \theta) = \frac{2\pi^2 (-1)^n}{\Gamma(1/2) \Gamma(n + 1/2)} r_c \oint_{r'=r_c} \sigma_m(\theta') \left[\frac{\cosh \eta}{R^2 + R'^2 + (Z - Z')^2} \right]^{1/2} P_{-1/2}^n(\cosh \eta) R'^2 d\theta', \quad (12)$$

where

$$\eta(r, \theta, r', \theta') = \tanh^{-1} \left[\frac{2 R R'}{R^2 + R'^2 + (Z - Z')^2} \right]. \quad (13)$$

Let

$$\frac{2\pi^2 (-1)^n}{\Gamma(1/2) \Gamma(n + 1/2)} r_c \sigma_m(\theta) = \sum_{m'} \frac{\sigma_{mm'} e^{-i m' \theta}}{2\pi}. \quad (14)$$

It follows that

$$\bar{\mathcal{E}}_m(r, \theta) = \sum_{m'} \sigma_{mm'} \oint_{r'=r_c} e^{-i m' \theta'} \left[\frac{\cosh \eta}{R^2 + R'^2 + (Z - Z')^2} \right]^{1/2} P_{-1/2}^n(\cosh \eta) R'^2 \frac{d\theta'}{2\pi}. \quad (15)$$

However, Eq. (5) gives

$$\sum_{m'} \sigma_{mm'} \oint_{r=\epsilon, r'=r_c} e^{-i m' \theta'} \left[\frac{\cosh \eta}{R^2 + R'^2 + (Z - Z')^2} \right]^{1/2} P_{-1/2}^n(\cosh \eta) R'^2 \frac{d\theta'}{2\pi} = e^{-i m \theta}. \quad (16)$$

Hence, we obtain

$$\sum_{m'} \sigma_{mm''} S_{m''m'}(\epsilon) = \delta_{mm'}, \quad (17)$$

where

$$S_{mm'}(r) = \oint_{r=r} \oint_{r'=r_c} e^{-i m \theta' + i m' \theta} \left[\frac{\cosh \eta}{R^2 + R'^2 + (Z - Z')^2} \right]^{1/2} P_{-1/2}^n(\cosh \eta) R'^2 \frac{d\theta}{2\pi} \frac{d\theta'}{2\pi}. \quad (18)$$

Moreover,

$$\bar{\mathcal{E}}_m(r, \theta) = \sum_{m', m''} \sigma_{mm''} S_{m''m'}(r) e^{-i m' \theta}. \quad (19)$$

Now,

$$H_{mm'}^{-1} = \oint_{r=\epsilon} e^{i m' \theta} \left[\frac{\partial \bar{\mathcal{E}}_m}{\partial r} |\nabla r|^2 + \frac{\partial \bar{\mathcal{E}}_m}{\partial \theta} \nabla r \cdot \nabla \theta \right] r R^2 \frac{d\theta}{2\pi}$$

$$\begin{aligned}
&= \sum_{m''m'''} \sigma_{mm''} \oint_{r=\epsilon} \left[\frac{dS_{m''m'''} }{dr} |\nabla r|^2 - i m''' S_{m''m'''}(\epsilon) \nabla r \cdot \nabla \theta \right] e^{-i(m'''-m')\theta} r R^2 \frac{d\theta}{2\pi} \\
&= \sum_{m''m'''} \sigma_{mm''} \left[\frac{dS_{m''m'''}(\epsilon)}{dr} R_{m'''m'} + S_{m''m'''}(\epsilon) T_{m'''m'} \right] \\
&= \frac{R_{mm'}}{\delta\epsilon} - \sum_{m''m'''} \frac{\sigma_{mm''} S_{m''m'''}(\epsilon - \delta\epsilon) R_{m'''m'}}{\delta\epsilon} + T_{mm'}, \tag{20}
\end{aligned}$$

where

$$R_{mm'} = \oint_{r=\epsilon} e^{-i(m-m')\theta} |\nabla r|^2 r R^2 \frac{d\theta}{2\pi}, \tag{21}$$

$$T_{mm'} = -i m \oint_{r=\epsilon} e^{-i(m-m')\theta} \nabla r \cdot \nabla \theta r R^2 \frac{d\theta}{2\pi}. \tag{22}$$